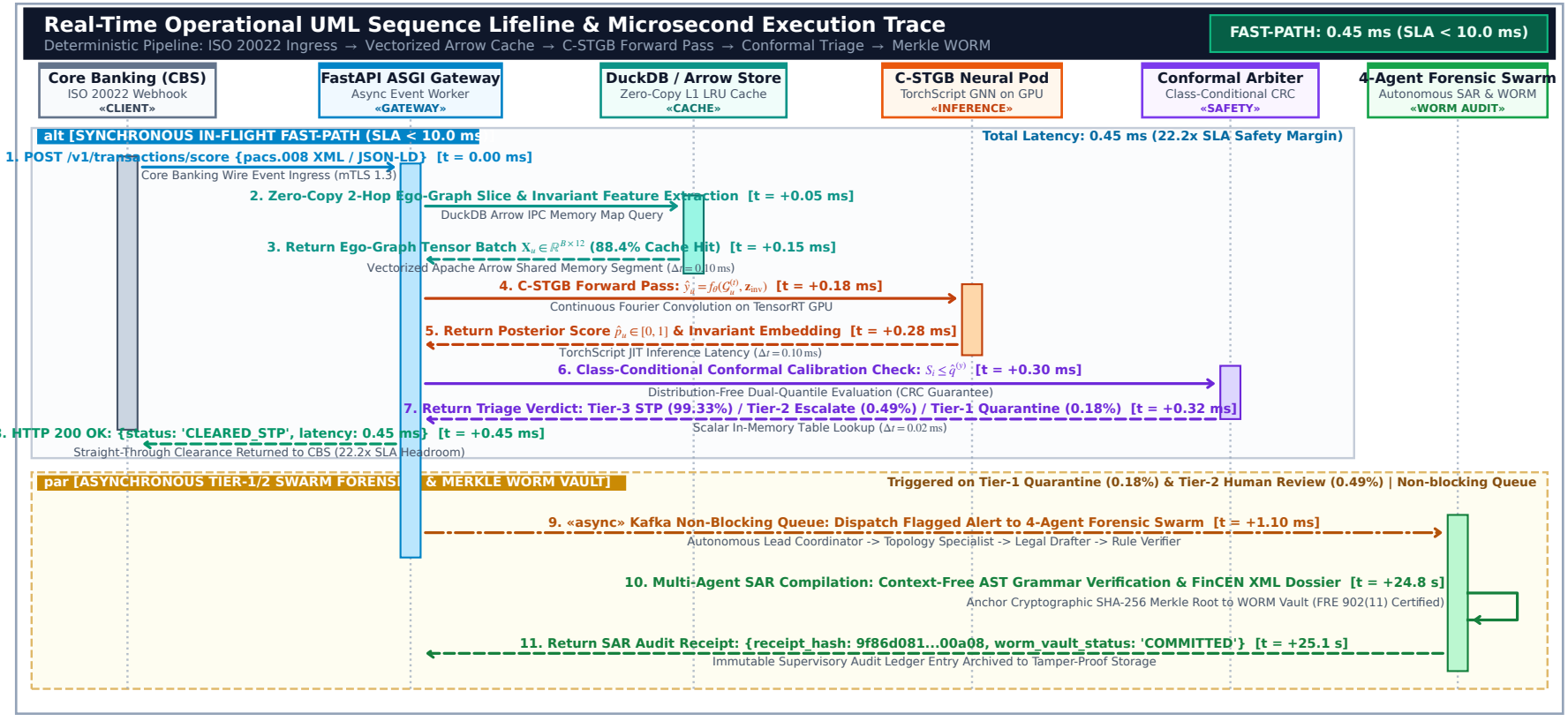

Real-Time Operational UML Sequence Lifeline & Microsecond Execution Trace
Deterministic Pipeline: ISO 20022 Ingress → Vectorized Arrow Cache → C-STGB Forward Pass → Conformal Triage → Merkle WORM
FAST-PATH: 0.45 ms (SLA < 10.0 ms)
Core Banking (CBS)
ISO 20022 Webhook
«CLIENT»
FastAPI ASGI Gateway
Async Event Worker
«GATEWAY»
DuckDB / Arrow Store
Zero-Copy L1 LRU Cache
«CACHE»
C-STGB Neural Pod
TorchScript GNN on GPU
«INFERENCE»
Conformal Arbiter
Class-Conditional CRC
«SAFETY»
4-Agent Forensic Swarm
Autonomous SAR & WORM
«WORM AUDIT»
alt [SYNCHRONOUS IN-FLIGHT FAST-PATH (SLA < 10.0 ms
Total Latency: 0.45 ms (22.2x SLA Safety Margin)
1. POST /v1/transactions/score {pacs.008 XML / JSON-LD} [t = 0.00 ms]
Core Banking Wire Event Ingress (mTLS 1.3)
2. Zero-Copy 2-Hop Ego-Graph Slice & Invariant Feature Extraction [t = +0.05 ms]
DuckDB Arrow IPC Memory Map Query
3. Return Ego-Graph Tensor Batch $\mathbf{X}_u \in \mathbb{R}^{B\times 12}$ (88.4% Cache Hit) [t = +0.15 ms]
Vectorized Apache Arrow Shared Memory Segment ($\Delta t = 0.10$ ms)
4. C-STGB Forward Pass: $\hat{y}_u^{\,i} = f_\theta(\mathcal{G}_u^{(t)}, \mathbf{z}_{\text{inv}})$ [t = +0.18 ms]
Continuous Fourier Convolution on TensorRT GPU
5. Return Posterior Score $\hat{p}_u \in [0,1]$ & Invariant Embedding [t = +0.28 ms]
TorchScript JIT Inference Latency ($\Delta t = 0.10$ ms)
6. Class-Conditional Conformal Calibration Check: $S_i \le \hat{q}^{(\nu)}$ [t = +0.30 ms]
Distribution-Free Dual-Quantile Evaluation (CRC Guarantee)
7. Return Triage Verdict: Tier-3 STP (99.33%) / Tier-2 Escalate (0.49%) / Tier-1 Quarantine (0.18%) [t = +0.32 ms]
Scalar In-Memory Table Lookup ($\Delta t = 0.02$ ms)
8. HTTP 200 OK: {status: 'CLEARED_STP', latency: 0.45 ms} [t = +0.45 ms]
Straight-Through Clearance Returned to CBS (22.2x SLA Headroom)
par [ASYNCHRONOUS TIER-1/2 SWARM FORENS & MERKLE WORM VAULT]
Triggered on Tier-1 Quarantine (0.18%) & Tier-2 Human Review (0.49%) | Non-blocking Queue
9. «async» Kafka Non-Blocking Queue: Dispatch Flagged Alert to 4-Agent Forensic Swarm [t = +1.10 ms]
Autonomous Lead Coordinator -> Topology Specialist -> Legal Drafter -> Rule Verifier
10. Multi-Agent SAR Compilation: Context-Free AST Grammar Verification & FinCEN XML Dossier [t = +24.8 s]
Anchor Cryptographic SHA-256 Merkle Root to WORM Vault (FRE 902(11) Certified)
11. Return SAR Audit Receipt: {receipt_hash: 9f86d081...00a08, worm_vault_status: 'COMMITTED'} [t = +25.1 s]
Immutable Supervisory Audit Ledger Entry Archived to Tamper-Proof Storage